Project 3 Report — Group10

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Team & Repo

- Group: 10
- Members: Muhan Zhang, Danhua Zhao
- Division of work: Muhan Zhang (2PC, Raft core, Docker), Danhua Zhao (testing, logs/screenshots, report)
- Base repo: `https://github.com/J1anXu/Distributed-picture-sharing-system`
- Submission repo (our repo): `https://github.com/ricolalemon/Distributed-picture-sharing-system-consensus`
- Consensus additions: `consensus_node/`, `docker-compose-consensus.yml`

Overview

Added a 5-node consensus cluster with Two-Phase Commit (vote+decision) and Raft (leader election, heartbeats, log replication). Followers forward client commands to the leader; majority commit persists state to `/data/consensus_store.json`.

Architecture Highlights

- Heartbeat: 1s; Election timeout: randomized 1.5–3s.
- Full-log replication on AppendEntries; majority ACK triggers commit.
- Protos: `consensus_node/two_phase.proto`, `consensus_node/raft.proto`.
- Docker: 5 homogeneous nodes; host ports 6001–6005 map to 6000.

Implementation Details

- 2PC: coordinator issues Vote to all peers; simple veto rule (keys starting with "deny" abort); pending txns tracked per node; Decision applies commit/abort and writes to KV store; required prints implemented on client/server.
- Raft: follower/candidate/leader roles; randomized election timeout to avoid split vote; leader heartbeats every 1s; full-log replication on AppendEntries; majority ACK advances commit index; client commands accepted on any node and forwarded to leader when needed.

- Persistence: per-node JSON at `/data/consensus_store.json` (Docker volume) for committed state.

Logging Formats (per requirements)

- 2PC (client): `Phase P of Node N sends RPC R to Phase P of Node M`
- 2PC (server): `Phase P of Node N runs RPC R called by Phase P of Node M`
- Raft (client): `Node X sends RPC <rpc> to Node Y`
- Raft (server): `Node X runs RPC <rpc> called by Node Y`

Container Topology

- Services: consensus-node1..5
- Ports: host 6001–6005 → container 6000
- Network: default docker network; all nodes reachable by service name
- Data: volume per node for `/data/consensus_store.json`

How to Run

1. Start:
   ```
   docker compose -f docker-compose-consensus.yml up --build -d
   ```
2. 2PC demo:
   ```
   grpcurl -plaintext -d '{"key":"k1","value":"v1"}'
   localhost:6001 twopc.TwoPhaseService/StartTransaction
   ```
3. Raft demo (follower OK):
   ```
   grpcurl -plaintext -d '{"command":"set a 1","client_id":"cli"}'
   localhost:6003 raft.RaftService/ClientCommand
   ```
4. Stop:
   ```
   docker compose -f docker-compose-consensus.yml down
   ```
5. Host helper:
   ```
   PYTHONPATH=. .venv-local/bin/python
   consensus_node/demo_raft.py "set a 1"
   ```
6. In container:
   ```
   PYTHONPATH=/app TARGET=consensus-nodeX:6000
   python consensus_node/demo_raft.py "set k v"
   ```

Validation (Q5 Test Cases)

1. Initial election: RequestVote + AppendEntries.
2. Heartbeat stability: steady AppendEntries, no elections.
3. Log replication: `set a 1` committed on all nodes.
4. Client forwarding: follower forwards ClientCommand to leader; commit visible on all nodes.
5. Leader failover/restart: stop old leader, elect new; restart old leader; post-failover `set c 3` committed everywhere.

Notes

- Compose omits version to silence warnings.
- Use `.venv-local/bin/python` on host; set `PYTHONPATH=/app` inside containers.
- Logs follow required 2PC and Raft RPC formats.
- If grpc missing: install deps with `.venv-local/bin/pip install -r requirements.txt`; rerun commands with `PYTHONPATH=. .venv-local/bin/python ...`
- Check state quickly: `docker exec <node> cat /data/consensus_store.json`

Screenshots

```
$ docker compose -f docker-compose-consensus.yml logs --tail 120
consensus-node4-1  | [consensus-node4] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node4-1  | Node consensus-node4 runs RPC RequestVote called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | [consensus-node5] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node5-1  | Node consensus-node5 runs RPC RequestVote called by Node consensus-node2
consensus-node1-1  | [consensus-node1] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC RequestVote called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | [consensus-node3] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node3-1  | Node consensus-node3 runs RPC RequestVote called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node2-1  | [consensus-node2] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
```

Figure 1: Initial election logs

```
$ docker compose -f docker-compose-consensus.yml logs --tail 60
consensus-node4-1  | [consensus-node4] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node4-1  | Node consensus-node4 runs RPC RequestVote called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | [consensus-node5] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node5-1  | Node consensus-node5 runs RPC RequestVote called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | [consensus-node1] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node1-1  | Node consensus-node1 runs RPC RequestVote called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node2-1  | [consensus-node2] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 runs RPC ClientCommand called by Node demo-client
consensus-node3-1  | [consensus-node3] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node3-1  | Node consensus-node3 runs RPC RequestVote called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC ClientCommand called by Node demo-client
consensus-node3-1  | Node consensus-node3 sends RPC ClientCommand to Node consensus-node2
```

Figure 2: Logs after set a 1

```
$ docker compose -f docker-compose-consensus.yml logs --tail 120
consensus-node3-1  | [consensus-node3] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node3-1  | Node consensus-node3 runs RPC RequestVote called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC ClientCommand called by Node demo-client
consensus-node3-1  | Node consensus-node3 sends RPC ClientCommand to Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | [consensus-node5] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node5-1  | Node consensus-node5 runs RPC RequestVote called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | [consensus-node4] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node4-1  | Node consensus-node4 runs RPC RequestVote called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC ClientCommand called by Node demo-client
consensus-node4-1  | Node consensus-node4 sends RPC ClientCommand to Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | [consensus-node1] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node1-1  | Node consensus-node1 runs RPC RequestVote called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node2-1  | [consensus-node2] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 runs RPC ClientCommand called by Node demo-client
consensus-node2-1  | Node consensus-node2 runs RPC ClientCommand called by Node demo-client
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
```

Figure 3: Leader failover logs

```
$ docker compose -f docker-compose-consensus.yml logs --tail 80
consensus-node3-1  | [consensus-node3] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node3-1  | Node consensus-node3 runs RPC RequestVote called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC ClientCommand called by Node demo-client
consensus-node3-1  | Node consensus-node3 sends RPC ClientCommand to Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node3-1  | Node consensus-node3 runs RPC AppendEntries called by Node consensus-node2
consensus-node2-1  | [consensus-node2] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node1
consensus-node4-1  | [consensus-node4] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC RequestVote to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node1-1  | [consensus-node1] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node5-1  | [consensus-node5] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node1-1  | Node consensus-node1 runs RPC RequestVote called by Node consensus-node2
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node5-1  | Node consensus-node5 runs RPC RequestVote called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC RequestVote called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | [consensus-node5] consensus node listening on 0.0.0.0:6000 with peers: ['consensus-node1', 'consensus-node2', 'consensus-node3', 'consensus-node4', 'consensus-node5']
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node4-1  | Node consensus-node4 runs RPC ClientCommand called by Node demo-client
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC AppendEntries called by Node consensus-node2
consensus-node5-1  | Node consensus-node5 runs RPC ClientCommand called by Node demo-client
consensus-node5-1  | Node consensus-node5 sends RPC ClientCommand to Node consensus-node2
consensus-node4-1  | Node consensus-node4 sends RPC ClientCommand to Node consensus-node2
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 runs RPC ClientCommand called by Node demo-client
consensus-node2-1  | Node consensus-node2 runs RPC ClientCommand called by Node demo-client
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node4-1  | Node consensus-node4 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node1-1  | Node consensus-node1 runs RPC AppendEntries called by Node consensus-node2
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node5
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node1
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node3
consensus-node2-1  | Node consensus-node2 sends RPC AppendEntries to Node consensus-node4
```

Figure 4: Leader restart logs

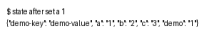

Figure 5: State after set a 1

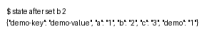

Figure 6: State after set b 2

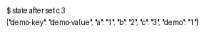

Figure 7: State after set c 3 (post-failover)